← All Roadmaps

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AI Engineer

Step by step guide to becoming an AI Engineer in 2026

Roadmap | Projects | AI Tutor | Personalize

Getting Started 4% | Halfway | Almost There | Complete

What is an AI Engineer?

Pre-requisites (One of these)
- Frontend
- Backend
- Full-Stack

Scrimba is offering 20% off on their AI Engineer course. Check them out!

Scrimba - AI Engineer Path

Related Roadmaps
- Vibe Coding
- Claude Code
- Prompt Engineering
- AI and Data Scientist Roadmap
- Forward Deployed Engineer Roadmap

Find the detailed version of this roadmap along with other similar roadmaps

roadmap.sh

AI Engineer

~~Introduction~~
- What is an AI Engineer?
- ~~Roles and Responsiblities~~
- ~~Impact on Product Development~~
- ~~AI Engineer vs ML Engineer~~

Working With LLMs

How LLMs Work

Core LLM Elements
- Tokens
- Context
- Sampling Parameters
- Temperature
- Top-K
- Top-P
- Repetition Penalties

Common Terminology
- ~~AI vs AGI~~
- Large Language Model (LLM)
- ~~Embeddings~~
- Training
- ~~Inference~~
- Vector DBs
- AI Agents
- RAGs
- Context Window
- Fine-tuning
- Prompt Engineering
- Context Engineering

Prompt Engineering

Prompt Engineering Roadmap

Prompt Anatomy
- Input Format
- System Prompting
- Role & Behavior
- Context
- Constraints
- Structured Output

Prompting Techniques
- Zero-Shot
- Few-Shot
- ReAct
- CoT

Model Interaction
- Function Calling
- Prompt Caching
- Streaming Responses

Context Engineering

Fundamentals
- Context vs Prompt Eng.
- What is a Context Layer?
- Context Sources
- MCP
- Context Security
- Context Evaluation

Context Eng. Techniques
- RAG & Dynamic Filters
- Memory Systems
- Context Compaction
- Long-Context Processing
- State & Historical Context
- Multi-agent Context Sharing
- Context Isolation
- Context Failure Modes

Tools & Context Warehouses
- modus
- DataHub
- Atlan
- PostHog

AI Models

Type of Models
- Pre-trained Models
- Closed vs Open Source Models
- Self-Hosted Models

Choosing the Right Model

Platforms & Ecosystem
- Hugging Face
- Hugging Face Tasks
- Hugging Face Hub
- Transformers.js
- Ollama
- LM Studio
- OpenRouter

APIs & SDKs
- OpenAI Response API
- Claude Messages API
- Google Gemini APi
- Hugging Face Inference SDK
- OpenAI-compatible APIs

Closed Models
- Anthropic Claude
- Google Gemini
- OpenAI (GPT, o-series)
- Cohere
- Mistral

Open Source Models
- Meta Llama
- DeepSeek
- Qwen
- Gemma

Embeddings & Vector Databases

What are Embeddings
- Semantic Search
- Data Classification
- Recommendation Systems
- Anomaly Detection

Use Cases for Embeddings

Embedding Models

Proprietary Models
- Open AI Embeddings API
- Gemini Embedding
- Cohere

Open Source Models
- Sentence Transformers
- Models on Hugging Face
- Jina

Vector Databases
- Purpose and Functionality

Popular Vector DBs (pick one)
- Chroma
- Pinecone
- Weaviate
- FAISS
- LanceDB
- Qdrant
- Supabase
- MongoDB Atlas

Implementing Vector Search
- Indexing Embeddings
- Performing Similarity Search

RAGs

What are RAGs?
- RAG Usecases
- RAG vs Fine-tuning

Implementing RAG
- Chunking
- Embedding
- Vector Database
- Retrieval Process
- Generation

Ways of Implementing RAG
- Using SDKs Directly
- Langchain
- Llama Index
- Haystack
- RAGFlow

AI Agents

AI Agents
- AI Agents Roadmap
- Agents Usecases
- ReAct Prompting
- Tools & Function Calling
- Multi-agents

Building AI Agents
- Manual Implementation
- OpenAI AgentKit & Agent SDK
- Claude Agent SDK
- Vertex AI Agent Builder
- Google ADK

Model Context Protocol (MCP)

Core Components
- MCP Host
- MCP Client
- MCP Server
- Data Layer
- Transport Layer

Developing wiht MCP
- Building an MCP Server
- Building an MCP Client
- Connect to Local Server
- Connect to Remote Server

AI Safety and Ethics

AI Safety and Ethics
- Prompt Injection Attacks
- Security and Privacy Concerns
- Bias and Fairness

Understanding AI Safety Issues
- Content Moderation APIs
- Adding end-user IDs in prompts
- Conducting adversarial testing
- Robust prompt engineering
- Know your Customers / Usecases
- Constraining outputs and inputs

Safety Best Practices

Evaluation & Observabiity

LLM Observability
- Tracing & logging
- Cost/latency monitoring
- Production monitoring

Observability Tools
- LangSmith
- Langfuse
- Helicone
- Arize AI

LLM Evaluations

Evaluation Types
- Deterministic Evals
- Model-Based Evals
- Human Evals
- Evaluation Metrics

Evaluation Tools
- DeepEval
- RAGAS

Regression Testing

Other AI Applications

Multimodal AI
- Multimodal AI Usecases
- Image Understanding
- Image Generation
- Video Understanding
- Audio Processing
- Text-to-Speech
- Speech-to-Text

Multimodal AI Tasks
- OpenAI Vision API
- DALL-E API
- NanoBanana API
- Whisper API
- Hugging Face Models
- LangChain for Multimodal Apps
- LlamaIndex for Multimodal Apps

Implementing Multimodal AI

Development Tools

AI Assisted Coding Tools
- Claude Code
- Gemini
- Codex
- Cursor
- Devin
- Replit
- Vibe Coding Roadmap

Continue learning with following relevant tracks
- Forward Deployed Eng.
- Prompt Engineering
- AI & Data Scientist

Frequently Asked Questions

Is AI engineering a good career choice?

Yes, becoming an AI engineer is a great career choice. AI is transforming nearly every industry, from healthcare and finance to transportation and entertainment. As companies increasingly adopt AI-driven solutions, the demand for skilled AI engineers continues to grow. This role offers opportunities to work on cutting-edge technologies like machine learning, natural language processing, and computer vision, making it a highly innovative field

What is reinforcement learning?

Do AI Engineers need a degree?

What is regression testing?

What is observability in LLM?

Join the Community

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367K	+3.2M	51K
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